TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	89.0 ± 1.18
Correlation Stability	86.3 ± 1.00
Distribution Consistency	92.8 ± 1.47
Entropy Preservation	90.5 ± 1.20
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	96.5 ± 1.36
Feature Accuracy	90.2 ± 1.66
Novelty Score	0.9 ± 0.01

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	52.666	+1.182	2.3%	OK
bp	76.469	77.688	+1.218	1.6%	OK
sg	1.017	1.018	+0.000	0.0%	OK
al	1.017	0.991	-0.026	2.6%	OK
su	0.450	0.494	+0.044	9.7%	OK
rbc	0.190	0.125	-0.065	34.0%	Drift
pc	0.227	0.203	-0.024	10.5%	Warn
pcc	0.106	0.122	+0.016	14.9%	Warn
ba	0.056	0.072	+0.016	29.4%	Drift
bgr	148.037	147.094	-0.943	0.6%	OK
bu	57.426	56.656	-0.770	1.3%	OK
sc	3.072	3.364	+0.292	9.5%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	89.0 ± 1.18
Correlation Stability (*2)	86.3 ± 1.00
Distribution Consistency (*3)	92.8 ± 1.47
Entropy Preservation (*4)	90.5 ± 1.20
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	96.5 ± 1.36
Feature Accuracy (*10)	90.2 ± 1.66
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9974999999999999	0.9724999999999999	0.99875	0.0012500000000000067	
F1	0.9977777777777778	0.9751235897586517	0.9990099009900991	0.0012321232123245	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	0.95625	0.9862500000000001	0.007500000000000007	
F1	0.9805471229310466	0.9595954169732565	0.9879433819362891	0.007396259005242467	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	0.9399999999999998	0.94625	-0.006249999999999989	
F1	0.9587663012688215	0.9467100659709994	0.9523592618291659	-0.006407039439655755	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9537500000000001	0.9425000000000001	0.9675	0.013750000000000007	
F1	0.9571570063241996	0.9497337432329906	0.9714450381036752	0.014288031779475685	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.83375	0.8450000000000001	0.6375	-0.19624999999999998	
F1	0.8685923018580157	0.8819002078331057	0.7150006018522587	-0.15359170000575692	*

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	0.9	0.9125	0.0025000000000000135	
F1	0.92384446063128	0.9119736801368354	0.925484890162355	0.00164042953107489 4	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9875	0.9487499999999999	0.99125	0.00374999999999999 77	
F1	0.9891384098753562	0.9533549442540977	0.9924019138755981	0.00326350400024174 6	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	0.6125	0.6575	0.045000000000000005	
F1	0.5885198185573248	0.6271068365587475	0.6708216937753336	0.08230187521800883	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	0.6	0.6024999999999999	0	
F1	0.7507214882964476	0.7483021334577380	0.7507214882964476	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	0.94625	0.9887500000000001	0.006250000000000011	
F1	0.9848619694663338	0.9496552284491196	0.9901039463221142	0.005241976855780372 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	82.0	80.0	1.015	1.0	0.0	0.0	0.0	0.0	1.0	309.0
2	56.0	80.0	1.01	4.0	2.0	0.0	1.0	1.0	1.0	140.0
3	63.0	70.0	1.01	3.0	4.0	0.0	1.0	1.0	1.0	163.0
4	17.0	100.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	99.0
5	67.0	90.0	1.01731316725978 66	3.0	0.0	0.0	1.0	1.0	0.0	150.0
6	70.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	119.0
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	65.0	60.0	1.025	0.0	0.0	0.0	0.0	1.0	0.0	111.0
2	63.0	80.0	1.01	1.0	1.0	0.0	0.0	0.0	0.0	140.0
3	28.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	117.0
4	42.0	140.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	119.0
5	54.0	70.0	1.015	4.0	0.0	0.0	0.0	0.0	0.0	122.0
6	65.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	97.0
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	64.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
2	45.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	93.0
3	58.0	90.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	238.0
4	75.0	90.0	1.01	3.0	1.0	0.0	1.0	1.0	0.0	106.0
5	37.0	110.0	1.01730902777777 8	2.0	0.0	0.0	1.0	1.0	0.0	101.0
6	45.0	100.0	1.01730902777777 8	4.0	0.0	0.0	0.0	0.0	1.0	92.0
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

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B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	71.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	106.0
2	21.0	80.0	1.015	4.0	0.0	0.0	1.0	0.0	1.0	165.0
3	54.0	70.0	1.01742907801418 44	3.0	0.0	0.0	0.0	0.0	0.0	380.0
4	29.0	140.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	360.0
5	60.0	80.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	220.0
6	43.0	80.0	1.025	0.0	3.0	0.0	0.0	0.0	0.0	123.0
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	55.0	80.0	1.02	2.0	0.0	0.0	0.0	0.0	0.0	102.0
2	76.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	70.0
3	70.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	121.0
4	40.0	90.0	1.025	2.0	0.0	0.0	0.0	0.0	1.0	93.0
5	30.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	79.0
6	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	131.0
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	69.0	80.0	1.01	0.0	0.0	0.0	0.0	0.0	0.0	123.0
2	71.0	60.0	1.02	0.0	3.0	0.0	0.0	0.0	0.0	93.0
3	62.0	60.0	1.01727915194346 27	4.0	0.0	1.0	0.0	1.0	0.0	230.0
4	48.0	100.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	114.0
5	68.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	101.0
6	74.0	70.0	1.01727915194346 27	1.0	3.0	0.0	0.0	0.0	0.0	307.0
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	22.0	70.0	1.015	1.0	0.0	0.0	0.0	0.0	0.0	172.0
2	72.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	93.0
3	42.0	100.0	1.01714788732394 37	2.0	0.0	0.0	0.0	0.0	0.0	146.0
4	65.0	70.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	117.0
5	46.0	80.0	1.01	0.0	0.0	0.0	0.0	0.0	0.0	490.0
6	44.0	80.0	1.01	0.0	0.0	0.0	0.0	0.0	0.0	121.0
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	33.0	70.0	1.01750877192982 46	2.0	0.0	1.0	1.0	0.0	1.0	140.0
2	39.0	100.0	1.015	0.0	2.0	0.0	0.0	0.0	0.0	121.0
3	71.0	100.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	117.0
4	47.0	70.0	1.015	4.0	2.0	0.0	0.0	0.0	1.0	139.0
5	34.0	80.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	121.0
6	44.0	70.0	1.025	1.0	0.0	0.0	1.0	1.0	0.0	219.0
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	40.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	121.0
2	61.0	70.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	118.0
3	55.0	60.0	1.01722807017543 86	0.0	0.0	0.0	1.0	1.0	0.0	108.0
4	55.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	99.0
5	46.0	90.0	1.02	1.0	0.0	0.0	0.0	0.0	0.0	230.0
6	52.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	91.0
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	63.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	121.0
2	36.0	70.0	1.025	1.0	2.0	0.0	1.0	1.0	0.0	184.0
3	45.0	60.0	1.01717081850533 8	1.0	0.0	1.0	0.0	0.0	0.0	463.0
4	45.0	90.0	1.015	4.0	0.0	0.0	0.0	0.0	0.0	121.0
5	72.0	90.0	1.01	1.0	4.0	0.0	0.0	0.0	0.0	156.0
6	47.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0